

TrES-2b

R-1833

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_160727 · created 2026-08-03T06:03:31+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457596.84924 +/- 0.00045 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.149 +/- 0.043
Transit depth (Rp/Rs)^2	2.22 +/- 1.29 [%]
Orbital Inclination (inc)	83.86 +/- 2.88
Airmass coefficient 1 (a1)	0.17 +/- 0.11
Airmass coefficient 2 (a2)	1.44 +/- 0.68
Scatter in the residuals of the lightcurve fit is	68.2 %
Best Comparison Star	#1 - [105, 456]
Optimal Aperture	3.02
Optimal Annulus	7.04
Transit Duration (day)	0.065 +/- 0.041

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.149 ± 0.043 vs 0.1254 (deviation 0.55σ)
Duration fit vs published	0.065 d vs 0.0746 d (deviation 0.23σ)
Mid-time O–C	0.3 min
Depth SNR	3.5
Residual scatter	68.2 %

Light curve

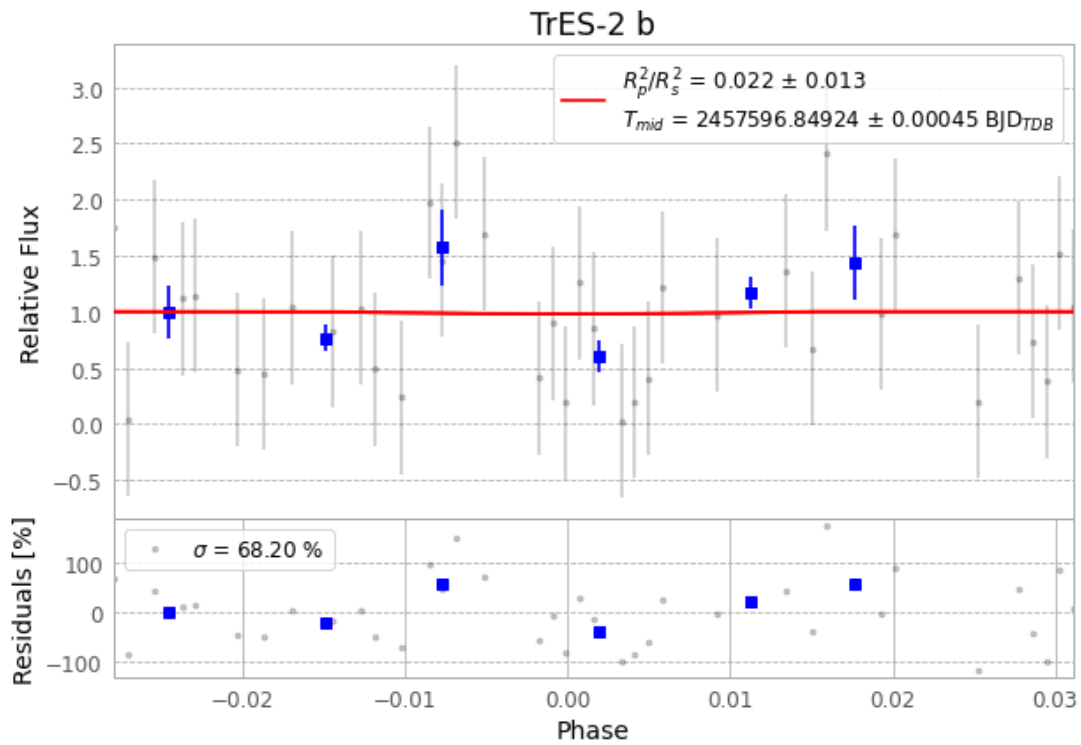


Figure 1 — FinalLightCurve_TrES-2 b_2016-07-26.png (SHA-256 a2467e5b609de502...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [273, 437], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 008aa79790c3d5e2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

71 FITS frames (47 MB), TRES-2160727063912.FITS → TRES-2160727100912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-160727.log (1ceb6060d01c...), FinalLightCurve_TrES-2 b_2016-07-26.png (a2467e5b609d...), FinalLightCurve_TrES-2 b_2016-07-26.csv (0c2f04b382fb...)

Compute resources

Wall time 170.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 06:03Z.